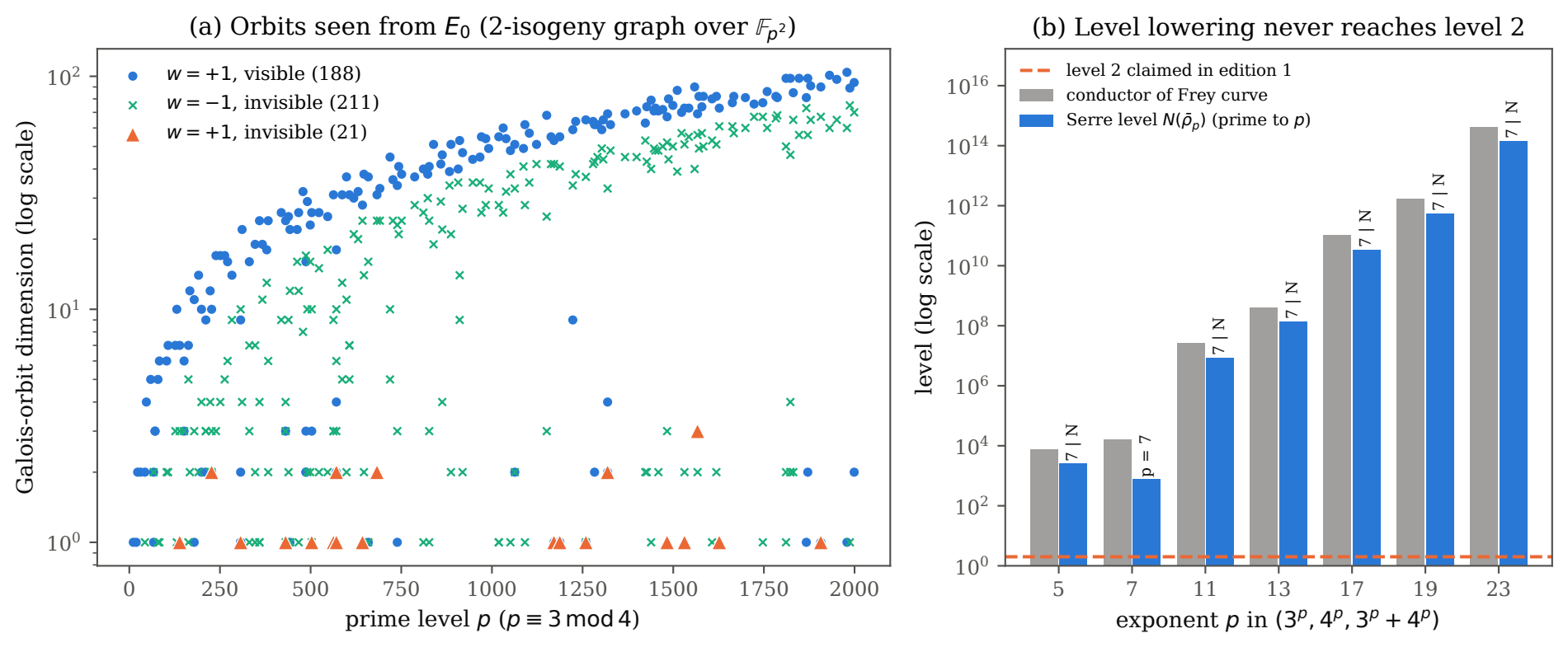
(a) Orbits seen from E_0 (2-isogeny graph over $\mathbb{F}_{p^2}$)
$w = +1$, visible (188)
$w = -1$, invisible (211)
$w = +1$, invisible (21)
Galois-orbit dimension (log scale)
10^2
10^1
10^0
0
250
500
750
1000
1250
1500
1750
2000
prime level p ($p \equiv 3 \bmod 4$)
(b) Level lowering never reaches level 2
level 2 claimed in edition 1
conductor of Frey curve
Serre level $N(\bar{\rho}_p)$ (prime to p)
level (log scale)
10^{16}
10^{14}
10^{12}
10^{10}
10^8
10^6
10^4
10^2
10^0
7 | N
p = 7
5
7
11
13
17
19
23
exponent p in $(3^p, 4^p, 3^p + 4^p)$